

Docker Container Restart Policy Issue SOP

Objective

Ensure containers restart automatically after failures, daemon restarts, or host reboots when the expected restart policy is missing or incorrect.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T028
Issue	Docker container restart policy issue
Category	Deployment
Service	Docker
Severity	Low
Status	Resolved
Description	Container not restarting on failure
Expected Resolution	Set restart policy in docker run

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
N/A	N/A	N/A	No direct log evidence found in logs CSV for this ticket.	N/A

Procedure

1. Confirm the container name, failure behavior, expected restart behavior, and deployment method.
2. Inspect the container restart policy using `docker inspect`.
3. Check whether the container was started with `docker run`, `compose`, or an orchestrated deployment.
4. Review exit code, logs, and whether the container stopped manually.
5. Set an approved restart policy such as `unless-stopped` or `on-failure` based on workload type.
6. Recreate or update the container with the correct restart policy.
7. Test by restarting the Docker daemon or simulating process failure in a safe environment.
8. Document the final restart policy in the ticket and deployment configuration.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Container not restarting	No restart policy	Set restart policy
Restarts only sometimes	on-failure max retry reached	Adjust retry count or policy
Does not restart after manual stop	Expected Docker behavior	Start manually or use unless-stopped
Compose ignores setting	Policy missing in compose file	Add restart key to service

Best Practices

1. Define restart policy in source-controlled deployment files.
2. Do not rely on manual docker run commands for critical services.
3. Match restart policy to workload type.
4. Monitor restart count and failure loops.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook